

Venkata Sai Sujith Palivela

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Summary

Computer Science undergraduate passionate about Artificial Intelligence and Machine Learning. Built a GST-focused Retrieval-Augmented Generation (RAG) system during an AI/ML internship at IIT Hyderabad.

Education

- Swarna Bharathi Institute of Science & Technology** Khammam, India
B.Tech in Computer Science and Engineering — CGPA: 8.7 2024 – Present
- S R Junior College** Telangana, India
Intermediate (Class XII) — Percentage: 84% 2022 – 2024
- V.V. Central Public School** India
Secondary School (Class X) — Percentage: 75% 2011 – 2022

Experience

- AI/ML Intern** May 2026 – July 2026
IIT Hyderabad Hyderabad, India
 - Designed a Retrieval-Augmented Generation (RAG) model for Question Answering on Legal Texts with a focus on Generating answers to queries based on GST.
 - Enforced semantic retrieval with ChromaDB, Sentence Transformers and Cross-Encoder reranking.
 - Built a Streamlit-based conversational interface with support for multiple LLMs.

Projects

- GST RAG Assistant**
Python | Streamlit | ChromaDB | Sentence Transformer | OpenRouter
 - Built an entire RAG application to retrieve answers to queries in GST Acts and Built
 - Improved retrieval by Cross-Encoder reranking and using Streamlit interface to interact with the retrieval model through chat.
 - Optimized and tested iteratively to achieve manual retrieval accuracy of 85%.
- SmartQueue — Healthcare App Architecture**
React Native | AI Symptom Triage | Real-Time Database
 - Engineered a high-performance cross-platform React Native application designed for real-time digital outpatient department (OPD) queue tracking and comprehensive clinic appointment booking systems.
 - Integrated an intelligent AI-based symptom classification module that parses patient inputs to automatically route cases to appropriate medical specialties, slashing manual triage overhead.
 - Designed and implemented a tamper-proof QR-code token framework enabling real-time queue position tracking, live processing updates, and push notifications for status updates.
- Smart Pill Dispenser System**
IoT Integration | Notification Engines | Workflow Logic
 - Designed a hardware-software workflow system that handles real-time scheduling logic and precise physical mechanisms to automatically release prescription medications based on user configurations.
 - Developed an asynchronous notification engine pushing instant SMS alerts and dedicated caregiver notifications to drastically minimize missed medication dosages and improve chronic care compliance.

Technical Skills and Interests

- Languages:** Python, JavaScript, SQL, HTML, CSS.
- AI/ML Architecture:** Retrieval-Augmented Generation (RAG), Sentence Transformers, Vector Embeddings, Machine Learning.
- Developer Tools & Platforms:** Git, GitHub, Supabase, Antigravity, Figma UI/UX Design.
- Areas of Interest:** Generative AI Systems, Full Stack Web Architecture.

Achievements & Hackathons

- Finalist at MumbaiHacks 2025** — Competed in the grand finale of a high-stakes hackathon focusing on building production-ready Agentic AI systems within the healthtech space.
- District-Level Ideathon Competitor** — Pitched innovative technical solutions and presented proofs-of-concept targeting structural workflows to regional evaluation panels.